

Question - Set A:

Suppose you have started a courier service, where customers can request delivery through available riders. Initially, there are 3 riders available. Design the **PathaoCourier** class in such a way that the given code produces the expected output. You are given the arrays **riderBalance** and **availableRider** to keep track of each rider's earnings and availability. For simplicity, assume that there will be only 3 riders in the service.

Tester Code	Output
<pre> public class PathaoCourierTester { public static void main(String[] args) { System.out.println("Total user: " + PathaoCourier.totalUsers); System.out.println("Total balance: " + PathaoCourier.totalBalance); System.out.println("1-----"); PathaoCourier p1 = new PathaoCourier("Alex"); PathaoCourier p2 = new PathaoCourier("Bob"); PathaoCourier p3 = new PathaoCourier("Carol"); PathaoCourier p4 = new PathaoCourier("Luffy"); System.out.println("2-----"); System.out.println(p1.deliveryRequest(250)); System.out.println(p2.deliveryRequest(300)); System.out.println(p3.deliveryRequest(200)); System.out.println(p4.deliveryRequest(400)); System.out.println("3-----"); PathaoCourier.showDetails(); System.out.println("4-----"); PathaoCourier.completeDelivery(p1); PathaoCourier.completeDelivery(p2); System.out.println(p4.deliveryRequest(400)); System.out.println(p1.deliveryRequest(350)); System.out.println(p2.deliveryRequest(470)); System.out.println("5-----"); PathaoCourier.showDetails(); System.out.println("6-----"); PathaoCourier.completeDelivery(p1); PathaoCourier.completeDelivery(p3); System.out.println("7-----"); System.out.println("Total user: " + PathaoCourier.totalUsers); System.out.println("Total balance: " + PathaoCourier.totalBalance); } } public class PathaoCourier { public static int[] riderBalance = {0, 0, 0}; public static boolean[] availableRider = {true, true, true}; public String name; // Your Code here </pre>	<pre> Total user: 0 Total balance: 0 1----- 2----- Rider 1 received Alex's request. Rider 2 received Bob's request. Rider 3 received Carol's request. No rider available. 3----- All rider delivery details: Rider 1 has collected 250 taka. Rider 2 has collected 300 taka. Rider 3 has collected 200 taka. 4----- Alex's delivery completed by Rider 1 Bob's delivery completed by Rider 2 Rider 1 received Luffy's request. Rider 2 received Alex's request. No rider available. 5----- All rider delivery details: Rider 1 has collected 650 taka. Rider 2 has collected 650 taka. Rider 3 has collected 200 taka. 6----- Alex's delivery completed by Rider 2 Carol's delivery completed by Rider 3 7----- Total user: 4 Total balance: 1500 </pre>

Question - Set B:

Suppose you have opened a small hotel where guests can check in to available rooms and check out after their stay. Initially, the hotel has 3 rooms. Design the **HotelBooking** class in such a way that the given code produces the expected output. You are given the arrays **roomRevenue** and **availableRoom** to keep track of each room's revenue and availability.

Tester Code	Output
<pre> public class HotelBookingTester { public static void main(String[] args) { System.out.println("Total guests: " + HotelBooking.totalGuests); System.out.println("Total revenue: " + HotelBooking.totalRevenue); System.out.println("1-----"); HotelBooking h1 = new HotelBooking("Alex"); HotelBooking h2 = new HotelBooking("Bob"); HotelBooking h3 = new HotelBooking("Carol"); HotelBooking h4 = new HotelBooking("Luffy"); System.out.println("2-----"); System.out.println(h1.checkIn(5000)); System.out.println(h2.checkIn(3500)); System.out.println(h3.checkIn(3000)); System.out.println(h4.checkIn(5000)); System.out.println("3-----"); HotelBooking.showDetails(); System.out.println("4-----"); HotelBooking.checkOut(h1); HotelBooking.checkOut(h2); System.out.println(h4.checkIn(8000)); System.out.println(h1.checkIn(4000)); System.out.println("5-----"); HotelBooking.showDetails(); System.out.println("6-----"); System.out.println("Total guests: " + HotelBooking.totalGuests); System.out.println("Total revenue: " + HotelBooking.totalRevenue); } } public class HotelBooking { public static int[] roomRevenue = {0, 0, 0}; public static boolean[] availableRoom = {true, true, true}; public String name; // Your Code here </pre>	<pre> Total guests: 0 Total revenue: 0 1----- 2----- Room 1 is booked by Alex. Room 2 is booked by Bob. Room 3 is booked by Carol. No room available. 3----- All room booking details: Room 1 has earned a total of 5000 taka. Room 2 has earned a total of 3500 taka. Room 3 has earned a total of 3000 taka. 4----- Alex checked out from Room 1. Bob checked out from Room 2. Room 1 is booked by Luffy. Room 2 is booked by Alex. 5----- All room booking details: Room 1 has earned a total of 13000 taka. Room 2 has earned a total of 7500 taka. Room 3 has earned a total of 3000 taka. 6----- Total guests: 4 Total revenue: 23500 </pre>

SET A

```
public class PathaoCourier {
    public static int totalUsers;
    public static int totalBalance;

    public static int[] riderBalance = {0, 0, 0};
    public static boolean[] availableRider = {true, true, true};

    public String name;
    public int riderNo;

    public PathaoCourier(String name) {
        this.name = name;
        totalUsers++;
    }

    public String deliveryRequest(int amount) {
        for (int i = 0; i < availableRider.length; i++) {
            if (availableRider[i]) {
                riderNo = i+1;
                riderBalance[i] += amount;
                totalBalance += amount;
                availableRider[i] = false;
                return("Rider " + riderNo + " received " + name + "'s request.");
            }
        }
        return("No rider available.");
    }

    public static void completeDelivery(PathaoCourier obj) {
        availableRider[obj.riderNo-1] = true;
        System.out.println(obj.name + "'s delivery completed by Rider " + obj.riderNo);
    }

    public static void showDetails() {
        System.out.println("All rider delivery details:");
        for (int i = 0; i < riderBalance.length; i++) {
            System.out.println("Rider " + (i+1)+ " has collected " + riderBalance[i] + "
taka.");
        }
    }
}
```

SET B

```
public class HotelBooking {
    public static int[] roomRevenue = {0, 0, 0};
    public static boolean[] availableRoom = {true, true, true};

    public static int totalGuests;
    public static int totalRevenue;

    public String name;
    public int roomNo = 0;

    public HotelBooking(String name) {
        this.name = name;
        totalGuests++;
    }

    public String checkIn(int amount) {
        for (int i = 0; i < availableRoom.length; i++) {
            if (availableRoom[i]) {
                roomNo = i + 1;
                roomRevenue[i] += amount;
                totalRevenue += amount;
                availableRoom[i] = false;
                return "Room " + roomNo + " is booked by " + name + ".";
            }
        }
        return "No room available.";
    }

    public static void checkOut(HotelBooking obj) {
        availableRoom[obj.roomNo - 1] = true;
        System.out.println(obj.name + " checked out from Room " + obj.roomNo + ".");
    }

    public static void showDetails() {
        System.out.println("All room booking details:");
        for (int i = 0; i < roomRevenue.length; i++) {
            System.out.println("Room " + (i + 1) + " has earned a total of " +roomRevenue[i] +
" taka.");
        }
    }
}
```

Rubric of Set A and Set B

Points	Marks
2 static variables & 1 instance variable(can create more) [1 Marks]	1
Constructor [1 Marks] 1) constructor declaration with parameter 2) Properly updating the variable	0.5 0.5
deliveryRequest() method [4 Marks] 1) method declaration with proper return type 2) return statement 3) Condition checking(availableRider/availableRoom) 4) Update static variable 5) Update instance variable	0.5 0.5 1 1 1
completeDelivery() [2 Marks] 1) method declaration with proper return type 2) Printing statement 3) Update static variable	0.5 0.5 1
showDetails() method [2 Marks]	2
Total	10